

Emotional Consistency for Fake News Detection

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Abstract

Fake news often uses emotionally charged language and exhibits inconsistent emotional shifts to influence audiences' viewpoints. Traditional fake news detection methods usually rely on textual content and emotion analysis, but often overlook the effects of emotional inconsistencies within narratives. In this research, we propose a framework that integrates emotional inconsistencies within narratives with textual features to improve the effectiveness of fake news detection. Our proposed model achieves an AUC of 0.8675 and an F1 score of 0.8909 on the FakeNewsNet dataset.

CCS Concepts

• Computing methodologies → Feature selection; Neural networks.

Keywords

Fake News Detection, Emotional Consistency, Emotional Intensity

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1 Introduction

Fake news has become a serious issue in this age of information, which can affect democratic institutions and erode public confidence in the media [1]. The spread of fake news can significantly shape public perception, particularly during major political events. One such example is the 2016 US presidential election, where the prevalent misinformation on social media affected the election [2]. Studies indicate that political fake news, especially on social media platforms such as X (previously Twitter), is often disseminated more rapidly and extensively than factual information [3]. The ramifications of fake news extend beyond the political arena. It has impacted financial markets and shaped significant economic decisions [4]. Misinformation during the COVID-19 pandemic also exacerbated public panic, impeded response efforts, and increased hesitancy in vaccination [5].

As social networks are becoming a primary source of news for many, fake news detection has become increasingly challenging due to the vast volume of user-generated content. The involvement of bots and trolls makes it difficult for traditional fact-checking

methods to keep pace. This is where machine learning and natural language processing (NLP) come in, as these technologies have the potential to automate the process of identifying fake news [6]. Recent research has shown that emotional signals can provide critical information on whether a news article is real or fake [3]. Many studies highlight that the integration of sentiment and emotion analysis into machine learning and deep learning models can improve the performance of fake news detection.

Although many approaches focus on detecting emotion within individual articles or headlines, research on examining emotional consistency across entire narratives is scarce. In fake news, it is common to have an abrupt emotional shift, disproportionate emotional intensity, or sudden changes in tone [7]. For instance, an article that initially presents neutral or positive information suddenly introduces alarming or emotionally intense content to attempt to influence the reader's perception or provoke an emotional response. Such patterns are often observed in fake news, where emotional manipulation is used to increase engagement and virality. In recent studies, Zhang et al. [8] propose a graph attention network-based model that combines both external knowledge comparison and sentiment analysis to classify fake news. Similarly, Cheng et al. [9] introduce a feature space for emotional consistency to capture the degree of emotional consistency between the modalities. However, research on understanding how emotional shifts evolve throughout a single article remains an unaddressed gap.

This study aims to fill this gap by introducing an "Emotional Consistency Score" to investigate the emotional trajectory of news items and to evaluate how the emotional tone changes throughout an article. Fake news articles often display inconsistent emotional tones, shifting from one emotion to another to manipulate readers, in contrast to traditional journalism, which maintains a more stable emotional tone [10]. Such an emotional inconsistency may signal possible manipulation or fabrication. The proposed system offers a framework for evaluating the authenticity of news by detecting emotion patterns that may suggest manipulation or deliberate distortion of information. We analyze the emotional trajectory of a news story by tracking how emotions, such as fear, anger, sadness, joy, trust, disgust, surprise, and anticipation, fluctuate throughout the story from beginning to end. We try to capture and quantify these emotional variations to uncover emotional cues that can signal the intent to emotionally manipulate or deceive the audience.

In addition, it contributes to the psychological aspects of fake news detection by showing how misinformation operates at the psychological level and the role that emotional appeal plays in shaping public perception. This analysis allows for the development of more resilient fact-checking methods and public awareness strategies, which can foster a more informed media environment.

The objective of this research is to improve the detection of fake news by integrating emotional features such as emotional intensity



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and emotional consistency with textual features. We analyze the emotional patterns within news articles and quantify their consistency to identify manipulative emotional cues that are often present in fake news. Our proposed model leverages contextual embeddings for text representation and concatenates these with emotional features extracted from the narrative. The combined representation is then passed through a predictive layer to classify the news as real or fake. We conducted experiments using a benchmark dataset FakeNewsNet. The results demonstrate that incorporating emotional information consistently improves performance over text-only models.

Our main contributions are: (1) proposing an emotion-enhanced text representation framework that captures the interplay between linguistic content and emotional context, (2) introducing an emotion scoring mechanism that quantifies emotional intensity across narratives, and (3) empirically validating that emotion-aware representations improve fake news detection accuracy on the selected dataset.

2 Related Work

Fake news detection systems can be broadly categorized into different types according to what features they use to build the detection model, for example, methods based on news content, methods based on social context and propagation, methods that utilize psychological features (e.g., emotion). This section reviews these approaches, focusing on DNN models that have become popular in recent years and the application of emotion analysis, ultimately highlighting the gap addressed by our proposed framework.

2.1 Recent Advances in Fake News Detection

Early approaches to fake news detection relied on traditional Machine Learning (ML) classifiers and stylometric features (e.g., lexical, syntax, grammar) to identify linguistic deception. However, the performance of these methods was often limited. In recent years, Deep Neural Networks (DNNs), especially graph-based models, have been widely used in fake news detection research. Beyond raw text, models have increasingly started to handle multimodal content (text, images, videos, etc.), for example, using Convolutional Neural Networks (CNNs) for visual features and combining them with text embeddings to detect contradictions between modalities [11]. Wang and Zhang [12] address feature extraction challenges using a dual discriminator for domain adversarial training to isolate invariant cross-modal features. In related work, Dessi et al. [13] demonstrate that LSTM models with suitable embeddings are effective for toxicity detection in online content moderation.

In addition to content-based methods, several studies leverage social context and propagation patterns for fake news detection. Graph-based models, such as Hierarchical Propagation Networks [14], capture repost cascades and user–post–source relationships to exploit temporal and structural diffusion signals. Shu et al. [15] emphasize the role of user behavior and source credibility.

2.2 Emotion-Aware Fake News Detection

In recent years, sentiment and emotion analysis has become an important topic of research on the detection of fake news, as many researchers find that disinformation frequently employs emotional

language to capture attention and spread rapidly. Vatsalan and Arachchilage [16] show that users’ emotional reactions to social media content can help identify disinformation, while Mahmud et al. [17] detect fake news using sentiment cues. Mackey et al. [18] further demonstrate that adding emotion vectors improves model performance. This recognition has made emotion-based detection a highly effective computational strategy. For example, Yadav and Gupta [19] use a transformer-based network to extract emotional features from text and multimodal data, and Jiang et al. [20] combine semantic and fine-grained emotional features via attention for high-accuracy classification. Kirishnni et al. [21] analyze emotional intensity and “temperature” to detect deliberate emotional manipulation in fake news. Emotion extraction for fake news detection relies on two primary methods: lexicon-based models and emotion embeddings. Lexicons can be used to extract explicit interpretable emotional categories [22], while emotion embeddings (learned through BERT or BiLSTM) capture nuanced context-dependent affective signals [8].

2.3 Emotional Consistency and Fake News

Fake news often employs sensationalist tactics to exaggerate, dramatize, or emotionally manipulate information to attract attention, provoke strong reactions, and increase engagement, often compromising accuracy or context [23]. When fake news contains mixed content (true and false), the reader can identify some components as true (which align with their preexisting knowledge or beliefs) and others as possibly false [24]. From a psychological perspective, emotionally inconsistent content can create cognitive dissonance [25] and make readers more susceptible to believing the content, even if it is contradictory or false. This type of fake news generation method capitalizes on emotional inconsistency to bypass critical thinking and appeal to the emotional state of the reader instead of their logical reasoning [26].

Traditional journalism and reliable news sources tend to maintain a more stable emotional tone throughout their reporting, especially in news articles that aim to inform rather than evoke a particular emotional response [27]. In contrast, fake news often uses inconsistent emotional tones to sway readers toward specific points of view. For example, an article that starts with a fearful tone, changes to a hopeful one, and ends with anger may try to guide the reader through an emotional journey that distorts the facts [10]. Such emotional inconsistency can act as a red flag, suggesting that the article does not aim to present an objective view, but rather to influence the reader’s emotions in a particular direction [28]. Therefore, we can gain insights into the potential authenticity of a text by analyzing its emotional consistency, as emotionally inconsistent articles are more likely to represent manipulative or misleading content.

This study aims to address the research gap mentioned above by introducing an emotion scoring system that analyzes emotional consistency throughout an entire news article. Using distance measures such as the Euclidean distance, cosine similarity, etc., this method provides a broader perspective on how emotion shifts within a news article. Our focus in this research is on sentence-level change, which will allow a more holistic understanding of how emotion can be manipulated throughout a fake news article.

3 Proposed Fake News Detection Model

We propose that emotions expressed in news articles, in whole or in individual sentences, may be crucial determinants in distinguishing between fake and true news. The proposed framework for identifying fake news comprises three modules: the emotion identification module, the embedding and feature extraction module, and the news classification module as shown in Figure 1.

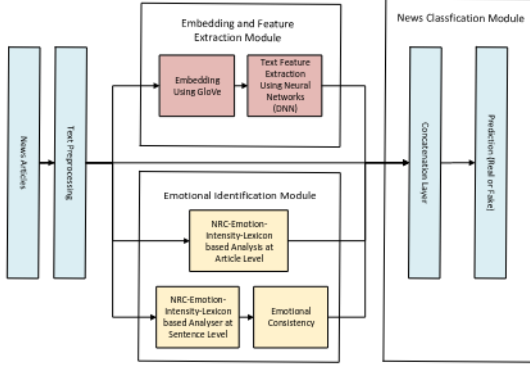


Figure 1: Proposed emotion-integrated framework.

3.1 Emotion Identification Module

This section focuses on how emotional cues are extracted and quantified in entire news articles. By examining the aggregated emotional intensity and patterns within an article, we aim to reveal emotional characteristics that can help distinguish between fake and real news.

3.1.1 Emotion Analysis at Article Level. The first step in this module is to detect the emotions present in each article. Most prior emotion-based fake news detection models focus on article-level emotions. Numerous studies have found that emotionally charged language is a distinguishing feature of misinformation [16, 17, 18]. Following these studies, we define and calculate the emotional intensity score to capture emotions at the article level. This method allows for a quantitative representation of the emotional content of an article (or any segment of text), which makes it suitable for further comparison and analysis. In this study, we apply a lexicon-based approach using the NRC-Emotion-Intensity-Lexicon [29]. This widely used resource associates words with different emotion categories, such as joy, sadness, anger, surprise, and fear, and captures their intensities.

Let A be an article consisting of words $w_1, w_2, \dots, w_n$. Each word w_i in the article is assigned an intensity score for each emotion e_j (e.g., joy, sadness, anger, etc.) using the NRC Emotion Intensity Lexicon.

$$I(A) = \sum_{i=1}^n \sum_{j=1}^8 S(w_i, e_j) \quad (1)$$

- $I(A)$: Total emotional intensity of article A ,
- n : Total number of words in article A ,
- w_i : The i^{th} word in article A ,
- e_j : The j^{th} emotion from the NRC Emotion Intensity Lexicon,

- There are 8 basic emotions in the lexicon: *anger, anticipation, fear, disgust, trust, surprise, joy, and sadness*,
- $S(w_i, e_j)$: The intensity score of word w_i for emotion e_j (0 if the word is not associated with that emotion).

The result is a single number that represents the emotional intensity score of the article. The emotion labels are typically binary (indicating the presence or absence of an emotion) or continuous (indicating the intensity of each emotion), depending on the lexicon and the analysis method. In this study, we consider continuous scores that represent the intensity of various emotions, allowing a more detailed representation of the emotional tone of each article [30]. To account for bias due to the length of the article, we normalize the total emotional intensity by dividing it by the number of words that trigger emotions, i.e. words that convey or evoke a specific emotional response, such as anger, joy, fear, or sadness, based on established emotion lexicons such as the NRC Emotion Lexicon.

3.1.2 Emotion Analysis at Sentence Level. The second step in this module involves detecting the emotions present in each sentence of the text. Again, in this study, we employ the NRC Emotion Lexicon-based approach [29].

For each sentence S , we calculate the intensity of each emotion by adding the individual intensity scores of the words present in the sentence according to the NRC lexicon. This is represented as:

$$I(S, e_j) = \sum_{i=1}^n S(w_i, e_j) \quad (2)$$

where:

- $I(S, e_j)$ is the intensity score for emotion e_j in sentence S ,
- n is the total number of words in the sentence,
- w_i represents the i^{th} word in the sentence,
- $S(w_i, e_j)$ is the intensity score of word w_i for emotion e_j according to the NRC lexicon,
- e_j represents the j^{th} emotion dimension (such as joy, sadness, anger, etc.).

After getting the intensity score for each emotion dimension, the sentence can then be represented as a vector of intensity scores, and the length of the vector is the number of emotion dimensions supported by the lexicon. This vector shows the emotional composition of the sentence.

3.1.3 Pairwise Comparison Between Sentences. Once emotion scores for all sentences have been obtained, the next step involves comparing the emotional differences between each pair of sentences in the article. Several distance measures are commonly used to calculate the dissimilarity or similarity between two documents in a data set. We apply these distance measures at the sentence level to capture and quantify subtle shifts in emotional intensity between sentences.

This approach helps identify emotional inconsistencies or patterns within an article that may not be apparent when using aggregated, document-level emotion scores. Consequently, it helps to assess how similar or dissimilar the emotional characteristics of different articles are, which is essential for detecting the emotional manipulation used in fake news. In the following paragraph, we provide a description of each distance measure and the reason why we choose it in our work.

Cosine Similarity provides a metric of similarity between two documents or feature vectors (emotion vectors in this case) by measuring the cosine of the angle between two vectors. *Euclidean Distance* is a straightforward and widely used measure for computing the distance between two points in a multidimensional space. *Jensen-Shannon Divergence (JSD)* is a smoothed and symmetrized version of the Kullback-Leibler divergence, which measures the divergence between two probability distributions [31]. Wasserstein distance is also a robust measure for comparing distributions by quantifying the minimum cost required to transform one distribution into another [32]. Bhattacharyya distance measures the similarity between two probability distributions [33], often used in classification and clustering problems.

3.1.4 Average of Emotional Differences. To assess the emotional consistency of a news article, it is essential to assess the emotional changes between sentences. Once the emotional differences between sentence pairs have been computed, we aggregate these differences to obtain a global measure of the emotional consistency of an article. This is crucial for understanding how emotions evolve throughout the text and whether these emotional shifts follow a coherent pattern or fluctuate erratically.

Let the emotional differences between sentence pairs be denoted as $\Delta E_{i,j}$, where i and j represent two sentences in the article. The average emotional difference, ΔE_{total} , is given by:

$$\Delta E_{total} = \sum_{i=1}^{N-1} \sum_{j=i+1}^N |\Delta E_{i,j}| \quad (3)$$

where:

- N is the total number of sentences in the article,
- $\Delta E_{i,j}$ is the emotional difference between sentence i and sentence j ,
- $|\Delta E_{i,j}|$ is the absolute emotional difference.

We normalize the emotional shifts (as shown in Equation (4)) by averaging the pairwise differences between sentences to ensure that it is not influenced by the length of the article. In the case of the cosine similarity measure, we do not subtract the average from 1, since cosine similarity directly represents the degree of similarity. Therefore, unlike other distance metrics where lower values indicate greater similarity, cosine similarity values closer to 1 signify higher similarity without requiring transformation.

$$\text{emo_con_score} = \begin{cases} 1 - \frac{\Delta E_{total}}{\text{num_pairs}} & \text{for distance measures} \\ \frac{\Delta E_{total}}{\text{num_pairs}} & \text{for cosine similarity} \end{cases} \quad (4)$$

The emotional consistency score provides valuable information on the emotional trajectory of a news article and offers a basis for distinguishing between fake and real news. The hypothesis underpinning this approach is that fake news articles tend to exhibit abnormal emotional fluctuations, where the emotional tone is often exaggerated or inconsistent with the overall narrative. These shifts can be used as early indicators of manipulation, helping to identify content designed to deceive or mislead readers.

The score is normalized to be within the range $[0,1]$, where a lower average value indicates greater emotional variance and suggests that the article exhibits erratic emotional transitions, which can potentially indicate manipulation, sensationalism, or intentional emotional fluctuations. A higher score, closer to 1, suggests that the article has a stable emotional tone, which is more characteristic of emotionally consistent narratives.

3.2 Embedding and Feature Extraction Module

This module extracts text-based features for the detection of fake news using neural network models. It begins with tokenizing and padding the processed news content. For word representation, we use pre-trained GloVe embeddings combined with a DNN model. GloVe embeddings map words into dense vectors, capturing both semantic and syntactic relationships across a corpus. While pre-trained models offer significant advantages in terms of reducing computational cost and training time, they may not fully capture the domain-specific nuances present in our dataset. To address this limitation and enhance the adaptability of the model to our specific task, we introduce an additional layer of DNN. This layer is designed to further refine and learn task-specific textual representations of the news articles and allows the model to better distinguish between real and fake news within the contextual boundaries of our dataset.

3.3 Classification Module

The classification module then combines these extracted text features with emotion-based features and applies fully connected layers to produce the final prediction.

3.3.1 Concatenation Layer. The Concatenation Layer merges textual feature representations extracted from models such as Bi-LSTM, Bi-GRU, MLP, CNN (which processes GloVe embeddings) with emotion-based features (emotional intensity and emotional consistency scores). The process involves the textual model, such as the Bi-LSTM, Bi-GRU, CNN and MLP, processing the input text to produce a feature vector, while the emotion identification module computes the scalar values for emotional intensity and consistency, forming another feature. The concatenate layer then takes these two vectors as input and generates a new combined feature vector. This fusion strategy enables subsequent dense layers to jointly learn from both the complex linguistic patterns and the emotional cues within a unified feature space, facilitating the capture of non-linear interactions between these complementary information sources for enhanced fake news detection.

3.3.2 Prediction Layer. The prediction layer is the final dense layer with a sigmoid activation function. It takes the output of the previous dense layer and produces a probability score, representing the likelihood that a given news article is fake or real. A sigmoid activation function is a standard choice for binary classification tasks, as it produces a probability value between 0 and 1, as defined in Equation 5, indicating the probability that a given piece of news belongs to a specific class (either “real” or “fake”).

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (5)$$

To optimize the model, the binary cross-entropy loss is minimized, which quantifies the difference between predicted probabilities and the true labels. Based on the output of the prediction layer, a class value (“real” or “fake”) is assigned.

4 Experiment

This section presents the experimental framework designed to assess the role of emotion features (emotional consistency and emotional intensity) in the detection of fake news. It is organized into four parts: experiment design, dataset details, evaluation metrics and system configuration, and hyperparameter fine-tuning. The results will be presented and discussed in the next section.

4.1 Experiment Design

The primary goal of our experiment is to evaluate the effectiveness of incorporating emotional features in improving the detection of fake news. There are mainly two groups of models that we tested in our experiment: 1) **Text-only Group**: uses only textual features without incorporating emotional consistency and emotional intensity. 2) **Emotion Integration Group**: integrates emotional features into the detection model, together with the textual features used in the text-only group.

4.2 Dataset

Since our model calculates emotional consistency by analyzing the emotional intensity at the sentence level in entire news articles, access to complete article texts is essential for our experiments. Identifying a suitable dataset to evaluate our proposed model is challenging, as most fake news datasets only contain the news headlines (without body texts) or short texts related to news, such as tweets or posts / comments (e.g., those extracted from subreddits), or they are not publicly available (e.g., NELA-GT). Following a thorough investigation and careful evaluation, we determine that FakeNewsNet [34] is the most suitable for our research. The news articles are scraped from the Internet and data are pre-processed for analysis. Articles containing fewer than two sentences are removed. The processed dataset contains 18,540 news articles, including 13,947 real articles and 4,593 fake ones. We conducted a preliminary analysis to compare the average emotional intensity and the average emotional consistency for fake and real news articles. The results indicate that, on average, fake news articles are more emotionally intense (0.8372) than real news (0.8153), suggesting that fake news tends to use stronger and more charged emotional language. For emotional consistency, when Euclidean distance is used, real news has slightly higher average consistency (0.5380) than fake news (0.5257).

Class imbalance is addressed using random oversampling for the minority class [35] and undersampling of the majority class [36]. Two additional columns are added to the dataset: the normalized emotional intensity scores of the news articles and the normalized emotional consistency scores based on the distance measures discussed in Section 3.1.3.

4.3 Evaluation Metrics & System Configuration

Model performance is evaluated using AUC (Area Under the Curve), Precision, Recall, and F1-score. All experiments are carried out on

a workstation equipped with a NVIDIA A100 80GB GPU, running Ubuntu 20.04 LTS, Python 3.11, and software libraries such as TensorFlow 2.19.0, scikit-learn 1.6.1, NumPy 1.26.4, imbalanced-learn 0.8.0, and nltk.

4.4 Parameter Selection

The parameter fine-tuning process is a critical component of any machine learning experiment. It typically involves the adjustment of the models’ parameters to align with the specific job being performed.

In our case, the following parameters are experimented with:

- **Batch size**: During training, we experimented with batch sizes of 128, 64, and 32 to balance computational efficiency and model performance.
- **Epochs**: We trained our models for 5 and 10 epochs to evaluate the impact of training duration on performance and convergence.
- **Learning rate**: We experimented with learning rates of 0.01, 0.001 and 0.0001 to optimize model convergence and performance.

Due to limited computational resources and time constraints, we did not employ a more exhaustive hyperparameter search, such as grid or random search. Instead, we evaluated a small set of configurations and identified that the optimal hyperparameter setting is 10 training epochs, a learning rate of 0.0001, and a batch size of 128.

5 Results

In this section, we present the experimental findings of our proposed fake news detection model. To address the issue of class imbalance in the dataset, we explored two different handling approaches: random undersampling of the majority class and random oversampling of the minority [36].

5.1 Results from Oversampling

The results obtained using the random oversampling technique demonstrate a clear improvement in the model’s ability to detect fake news. Table 1 shows how the performance of the model improves when emotional features are integrated with textual features. We have tested multiple DNN models in the embedding modules and multiple distance measures in the emotion consistency calculation. Here, we only show the results from CNN model and when the Euclidean distance is used. Similar patterns are observed when other DNN models and other distance measures are used.

As we can see in Table 1, the text-only model achieves an AUC of 0.8534 and a good balance between precision (0.8911) and recall (0.8585). These results indicate that textual features could capture linguistic signals, but may overlook deeper psychological signals that are often present in fake news[37]. When emotional intensity is introduced, the AUC and recall improve to 0.8662 and 0.8795, respectively. These results suggest that emotionally charged language in fake news offers additional discriminative power [3]. Incorporating emotional consistency further improves performance. Compared to the text-only model, the addition of emotional consistency yields higher AUC, F1-score, and recall. Compared to adding emotional intensity alone, the addition of emotional consistency further improves recall, contributing to an overall increase in the F1-score.

When both emotional features are included (emotional intensity and consistency), the AUC, recall, and F1 score reach the highest level. A decline in precision is observed, which reflects a trade-off between increased sensitivity and a higher false positive rate. In general, these results confirm that methods that incorporate emotional features have better performance than those that rely only on text information. Emotional cues at the narrative level complement traditional textual features at the surface level [38, 3].

Table 1: Performance comparison of different feature sets using CNN with Euclidean distance (oversampling).

Features Used	AUC	F1-score	Precision	Recall
Text only	0.8534	0.8729	0.8911	0.8585
Text + Emotional Intensity	0.8662	0.8826	0.8903	0.8795
Text + Emotional Consistency	0.8656	0.8857	0.8870	0.8919
Text + Intensity + Consistency	0.8675	0.8909	0.8827	0.9038

Table 2 presents a comparative evaluation of various deep neural network (DNN) architectures used in the embedding and feature extraction module. Two feature configurations are considered: (1) textual features only and (2) a combined feature set that incorporates textual and emotional features, using Euclidean distance for emotional consistency calculation. The results indicate that the inclusion of emotional intensity and consistency features enhances performance across all models. We observe the most notable improvement with the CNN model, which achieves the highest AUC of 0.8675 and an F1 score of 0.8909 when emotional features are incorporated. This suggests that the CNN model effectively leverages the added emotional cues to enhance class discrimination in the given task, due to their superior learning capacity.

Similarly, the Bi-LSTM architecture also benefits from emotional features, showing an increase in AUC and an increase in the F1 score. Its precision and recall also improve, indicating a more balanced performance in terms of true positive identification and the reduction of false negatives. The Bi-GRU model demonstrates performance gain with the addition of emotional features, with an increase in AUC from 0.8159 to 0.8175 but a small decrease in the F1 score from 0.8533 to 0.8513. This variability may be attributed to its simpler gating structure compared to Bi-LSTM, which may limit its ability to capture complex emotional patterns effectively. The MLP model also shows consistent improvements in both the AUC and the F1 score. Its recall increases. However, a slight decline in precision reflects a trade-off between detecting more positive instances and an increase in false positives.

Overall, the results suggest that regardless of the underlying DNN model, the integration of emotional features consistently improves performance in terms of AUC over the use of textual features alone. This consistency indicates that our proposed framework generalizes well and is not dependent on a specific neural architecture.

Table 3 shows the comparison of the results using different distance measures in the calculation of emotional consistency. This table only includes the results when the CNN model is used for text representation. Similar patterns can be observed when using other DNN models. All distance measures, including Cosine, Euclidean distance (EuclDis), Wasserstein distance, Jensen-Shannon divergence (JSD) and Bhattacharyya (Bhatt), consistently outperform

Table 2: Performance of different DNN architectures with and without emotional features using Euclidean distance (oversampling).

DNN	Feature	AUC	F1-score	Precision	Recall
Bi-LSTM	Text Only	0.8304	0.8562	0.8819	0.8292
	Text + Intensity + Consistency	0.8373	0.8674	0.8836	0.8505
Bi-GRU	Text Only	0.8159	0.8533	0.8746	0.8346
	Text + Intensity + Consistency	0.8175	0.8513	0.8734	0.8357
MLP	Text Only	0.7882	0.8701	0.8451	0.9081
	Text + Intensity + Consistency	0.8012	0.8734	0.8420	0.9160
CNN	Text Only	0.8534	0.8729	0.8911	0.8585
	Text + Intensity + Consistency	0.8675	0.8909	0.8827	0.9038

the case when only textual features are used (AUC = 0.8534). This suggests that emotional features play a vital role in distinguishing fake from real news.

Table 3: Performance of CNN with different distance measures (oversampling).

Feature	AUC	F1-score	Precision	Recall
Text Only	0.8534	0.8729	0.8911	0.8585
Text + Intensity + Consistency (Cosine)	0.8679	0.8865	0.8819	0.8959
Text + Intensity + Consistency (EuclDis)	0.8675	0.8909	0.8827	0.9038
Text + Intensity + Consistency (JSD)	0.8680	0.8875	0.8916	0.8829
Text + Intensity + Consistency (Wass)	0.8673	0.8889	0.8899	0.8886
Text + Intensity + Consistency (Bhatt)	0.8669	0.8908	0.8880	0.8944

Among the distance measures tested, as we can see, the Euclidean distance achieves the highest F1 score (0.8909), demonstrating its strong overall classification performance. Its highest recall (0.9038) indicates effectiveness in identifying real instances, but possibly at the cost of precision. Jensen-Shannon divergence (JSD) achieves the highest AUC score (0.8680), which indicates its strength to capture the divergence in emotion distributions between sentences. Its probabilistic and symmetric nature likely makes it more robust in comparing soft probability distributions such as emotional vectors. In addition, it achieves the best precision (0.8916), which demonstrates its ability to reduce false positives. Cosine similarity performs competitively in AUC (the second highest), suggesting its effectiveness in capturing directional similarity between emotional vectors of sentences. Its recall (0.8959) is also the second highest, just behind the Euclidean distance, maintaining a good balance across all metrics. The Bhattacharyya distance shows a strong performance in the F1 score, achieving the second highest score (0.8909). Their performance further underscores that subtle shifts in emotion distributions carry discriminative signals for classification.

5.2 Results from Undersampling

The superior results observed with oversampling may stem from overfitting, where repeated minority samples lead the model to memorize those instances rather than generalize [39]. To further investigate the effect of different ways of handling class imbalance, we apply the random undersampling approach. Unlike oversampling, which increases minority class representation, undersampling balances the dataset by reducing the majority class. The results of this approach are presented in Tables 4 and 5. The results demonstrate

that the inclusion of emotional features consistently improves performance, although the gains are comparatively smaller than the oversampling (a possible reason is the reduced dataset size).

For CNNs, adding emotional intensity and consistency features provides the best results, achieving an AUC of 0.8667, an F1 score of 0.8933, and a recall of 0.9031, outperforming the case when only text is used. Similarly, Bi-LSTM, Bi-GRU and MLP models also benefit from emotional features.

Table 4: Performance comparison of different feature sets using CNN and Euclidean distance (undersampling).

Features Used	AUC	F1-score	Precision	Recall
Text only	0.8230	0.8665	0.8798	0.8573
Text + Emotional Intensity	0.8429	0.8669	0.8887	0.8548
Text + Emotional Consistency	0.8522	0.8797	0.8880	0.8768
Text + Intensity + Consistency	0.8667	0.8933	0.8868	0.9031

Table 5: Performance of different DNN architectures with and without Emotional Intensity and Consistency features using Euclidean distance (undersampling).

DNN	Feature	AUC	F1-score	Precision	Recall
Bi-LSTM	Text Only	0.8166	0.8398	0.8772	0.8162
	Text + Intensity + Consistency	0.8190	0.8476	0.8808	0.8202
Bi-GRU	Text Only	0.7821	0.8348	0.8528	0.8350
	Text + Intensity + Consistency	0.7896	0.8407	0.8665	0.8234
MLP	Text Only	0.7972	0.8161	0.8748	0.7733
	Text + Intensity + Consistency	0.8007	0.8285	0.8759	0.7960
CNN	Text Only	0.8230	0.8665	0.8798	0.8573
	Text + Intensity + Consistency	0.8667	0.8933	0.8868	0.9031

5.3 Ablation Study

To evaluate the necessity of adding a DNN layer after the pre-trained embedding model, we perform an ablation study to compare the model performance with and without any neural network architecture. In this study, we consider the model using both emotional and textual features. As shown in Table 6, the addition of a DNN layer significantly improves performance over using pre-trained GloVe embeddings alone. This improvement is consistent across all the DNNs evaluated, demonstrating that incorporating a DNN leads to richer, task-specific representations. Among DNNs, CNN shows the largest relative performance gain, with an improvement of 17.02% compared to using GloVe embeddings alone. This improvement can be attributed to CNNs' ability to effectively capture local semantic and emotional patterns across the text. These features are particularly important for detecting emotionally charged or misleading narratives, as they enable the model to learn more discriminative and context-aware representations to enhance classification performance. The results shown in the table are generated using the Euclidean distance. We observe similar patterns when other distance measures are used.

Table 6: Performance comparison of with and without DNN layer (multiple DNN architectures) using Euclidean distance as the distance measure.

DNN Model	AUC	F1-score	Precision	Recall
w/o DNN	0.7413	0.7530	0.8709	0.6757
Bi-LSTM	0.8373	0.8674	0.8836	0.8505
Bi-GRU	0.8175	0.8513	0.8734	0.8357
MLP	0.8012	0.8734	0.8420	0.9160
CNN	0.8675	0.8909	0.8827	0.9038

5.4 Discussion

The main purpose of our proposed model and the experiment design is to show that adding fine-grained emotional features such as emotional consistency can improve the fake news detection results. The way we design the emotion identification module is to make it a supplementary module that can be easily integrated with any existing detection model. This is one of the main reasons we did not compare our model with other fake news detection models in the experiment. In fact, the whole framework is designed in a modular way. This flexibility allows future researchers and practitioners to experiment with other emotional features or other types of feature, other ways of representing text, and other ways of fusion.

Among the research works that considered emotions or sentiment in the fake news detection model and used the FakeNewsNet dataset in the experiment, the studies by Ding et al. [40] and Shaeri and Katanforoush [41] are the closest to our work. However, since the authors did not make their code publicly available, we were unable to perform a direct experimental comparison. Based on the results (precision 0.73, recall 0.68 from [40] and precision, recall, and F1 of 0.87, 0.86, 0.87 from [41]) reported in their papers, our proposed model demonstrates better performance.

6 Conclusion and Future Work

To identify fake news, a framework based on emotional intensity and emotional consistency is proposed. Our framework integrates emotional cues at the article and sentence levels to improve fake news detection, we evaluate it using the FakeNewsNet dataset with random oversampling and undersampling to address class imbalance. Experimental results demonstrate that the incorporation of emotional features significantly improves the classification performance. Among these features, emotional consistency, quantified through various distance measures between emotion distributions throughout the narrative, emerges as particularly useful.

Despite these encouraging results, several limitations should be noted. First, oversampling and undersampling may have influenced the stability and robustness of the results. Second, we evaluated model performance on a single dataset (FakeNewsNet), and its findings may not be generalizable. Future research should validate these results across diverse datasets in fake news domains. Another important direction for future research is to examine how emotional inconsistency influences user perception and credibility judgments. Last but not least, the proposed method for calculating emotional consistency considers all pairs of sentences within the text. An alternative approach is to limit this analysis to only adjacent sentence pairs, focusing specifically on the emotional change from

one sentence to the next. This focused analysis could potentially provide new insight and improve fake news detection methods.

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References

- [1] Xinyi Zhou and Reza Zafarani. 2021. A survey of fake news: fundamental theories, detection methods, and opportunities. *ACM Computing Surveys*, 53, (Sept. 2021), 1–40. <https://dl.acm.org/doi/10.1145/3395046>.
- [2] Nir Grinberg, Kenneth Joseph, Lisa Friedland, Briony Swire-Thompson, and David Lazer. 2019. Fake news on twitter during the 2016 u.s. presidential election. *Science*, 363, 6425, 374–378. doi:10.1126/science.aau2706.
- [3] Soroush Vosoughi, Deb Roy, and Sinan Aral. 2018. The spread of true and false news online. *Science*, 359, (Mar. 2018), 1146–1151. <https://www.science.org/doi/10.1126/science.aap9559>.
- [4] Hunt Allcott and Matthew Gentzkow. 2017. Social media and fake news in the 2016 election. *Journal of Economic Perspectives*, 31, 2, 211–236. <https://pubs.aea.org/doi/10.1257/jep.31.2.211>.
- [5] Maria Mercedes Ferreira Caceres et al. 2022. The impact of misinformation on the COVID-19 pandemic. *AIMS Public Health*, 9, 2, 262–277. <https://doi.org/10.3934/publichealth.2022018>.
- [6] Jingfeng Cui, Zhaoxia Wang, Seng-Beng. Ho, and Erik Cambria. 2023. Survey on sentiment analysis: evolution of research methods and topics. *Artificial Intelligence Review*, 56, 8, 8469–8510. <https://doi.org/10.1007/s10462-022-10386-z>.
- [7] Renatha Souza Vieira and Álvaro Figueira. 2025. Emotional sequencing as a marker of manipulation in social media disinformation. *Future Internet*, 17, 12. doi:10.3390/fi17120546.
- [8] Hao Zhang, Zonglin Li, Sanya Liu, Tao Huang, Zhouwei Ni, Jian Zhang, and Zhihan Lv. 2024. Do sentence-level sentiment interactions matter? sentiment mixed heterogeneous network for fake news detection. *IEEE Transactions On Computational Social Systems*, 11, (Aug. 2024), 5090–5100. <https://ieeexplore.ieee.org/document/10123412/>.
- [9] Haonan Cheng, Zhenjia Shi, Xiang Lei, and Long Ye. 2023. CACEE: computational aesthetic classification of expressive effects based on emotional consistency. In *Proceedings of the 4th International Workshop on Human-centric Multimedia Analysis*. (Nov. 2023), 27–32. <https://dl.acm.org/doi/10.1145/360604.13618061>.
- [10] Bilal Ghanem, Paolo Rosso, and Francisco Rangel. 2020. An emotional analysis of false information in social media and news articles. *ACM Transactions on Internet Technology*, 20, 2, (May 2020), 1–18. doi:10.1145/3381750.
- [11] Yang Wu, Pengwei Zhan, Tianyang Liu, and Zhen Xu. 2021. Multimodal Fusion with Co-Attention Networks for Fake News Detection. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 2556–2566. doi:10.18653/v1/2021.findings-acl.226.
- [12] Bofan Wang and Shenwu Zhang. 2024. A multi-reading habits fusion adversarial network for multi-modal fake news detection. *International Journal of Advanced Computer Science and Applications*, 15, 7. doi:10.14569/IJACSA.2024.0150740.
- [13] Danilo Dessì, Diego Reforgiato Recupero, and Harald Sack. 2021. An assessment of deep learning models and word embeddings for toxicity detection within online textual comments. *Electronics*, 10, 7, (Mar. 25, 2021), 779. doi:10.3390/electronics10070779.
- [14] Van-Hoang Nguyen, Kazunari Sugiyama, Preslav Nakov, and Min-Yen Kan. 2020. FANG: leveraging social context for fake news detection using graph representation. In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management*. doi:10.1145/3340531.3412046.
- [15] Kai Shu, Suhang Wang, and Huan Liu. 2019. Beyond news contents: the role of social context for fake news detection. In *Twelfth ACM International Conference on Web Search and Data Mining (WSDM '19)*. (Jan. 2019), 312–320. <https://dl.acm.org/doi/10.1145/3289600.3290994>.
- [16] Dinusha Vatsalan and Nalin Asanka Gamagedara Arachchilage. 2021. Hold on! your emotion and behaviour when falling for fake news in social media. *Preprints*, (May 20, 2021). <https://www.preprints.org/manuscript/202105.0477/v1>.
- [17] Yuzi Mahmud, Noor Sakinah Shaeali, and Sofianita Mutalib. 2021. Comparison of machine learning algorithms for sentiment classification on fake news detection. *International Journal of Advanced Computer Science and Applications*, 12, 10. doi:10.14569/IJACSA.2021.0121072.
- [18] Andrew L. Mackey, Susan Gauch, and Kevin Labille. 2021. Detecting fake news through emotion analysis. In *Thirteenth International Conference on Information, Process, and Knowledge Management*. Nice, France, (July 2021).
- [19] Ashima Yadav and Anika Gupta. 2024. An emotion-driven, transformer-based network for multimodal fake news detection. *International Journal of Multimedia Information Retrieval*, 13, 1, (Mar. 2024), 7. doi:10.1007/s13735-023-00315-3.
- [20] Wen Jiang, Mingshu Zhang, Wei Bin, Xiong Zhang, Kelan Ren, and Facheng Yan. 2024. A model for detecting fake news by integrating domain-specific emotional and semantic features. *Computers, Materials & Continua*, 80, 2, 2161–2179. doi:10.32604/cmc.2024.053762.
- [21] Kirishnmi Prabagar, Kogul Srikanthabala, Nilaan Loganathan, Shalinka Jayatilake, Gihan Gamage, and Daswin De Silva. 2025. Addressing the information asymmetry of fake news detection using large language models and emotion embeddings. *Symmetry*, 17, 8, 1290. Open Access. doi:10.3390/sym17081290.
- [22] Fei Liu, Xinsheng Zhang, and Qi Liu. 2024. An emotion-aware approach for fake news detection. *IEEE Transactions on Computational Social Systems*, 11, 3, 3516–3524. doi:10.1109/TCSS.2023.3335269.
- [23] David M. J. Lazer et al. 2018. The science of fake news. *Science*, 359, (Mar. 2018), 1094–1096. <https://www.science.org/doi/10.1126/science.aao2998>.
- [24] Lionel Z. Wang, Yiming Ma, Renfei Gao, Beichen Guo, Han Zhu, Wenqi Fan, Zexin Lu, and Ka Chung Ng. 2024. MegaFake: a theory-driven dataset of fake news generated by large language models. *arXiv preprint*, (Aug. 2024). <https://arxiv.org/abs/2408.11871>.
- [25] Amanda S. Hinojosa, William L. Gardner, H. Jack Walker, Claudia Coglisier, and Daniel Gullifor. 2017. A review of cognitive dissonance theory in management research: opportunities for further development. *Journal of Management*, 43, 1, 170–199. doi:10.1177/0149206316668236.
- [26] Leon Festinger. 1957. *A Theory of Cognitive Dissonance*. Stanford University Press.
- [27] Yelena Mejova, Amy X. Zhang, Nicholas Diakopoulos, and Carlos Castillo. 2014. Controversy and sentiment in online news. *arXiv preprint arXiv:1409.8152*. <https://doi.org/10.48550/arXiv.1409.8152>.
- [28] Edson C. Tandoc, Zheng Wei Lim, and Richard Ling. 2017. Defining “fake news”: a typology of scholarly definitions. *Digital Journalism*, 6, 2, 137–153. <https://doi.org/10.1080/21670811.2017.1360143>.
- [29] Saif Mohammad. 2017. Word affect intensities. *arXiv preprint*, (2017). <https://arxiv.org/abs/1704.08798>.
- [30] Sven Buechel and Udo Hahn. 2017. EmoBank: studying the impact of annotation perspective and representation format on dimensional emotion analysis. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*. (Apr. 2017), 578–585. <https://aclanthology.org/E17-2092/>.
- [31] Jhoan K. Hoyos-Osorio and Luis G. Sanchez-Giraldo. 2023. The representation jensen-shannon divergence. Version Number: 4. (May 25, 2023). arXiv: 2305.16446. doi:10.48550/arXiv.2305.16446.
- [32] Aiming Zhang, Lei Su, Yin Zhang, Yunfa Fu, Liping Wu, and Shengjin Liang. 2021. EEG data augmentation for emotion recognition with a multiple generator conditional wasserstein GAN. *Complex & Intelligent Systems*, 8, 4, 3059–3071. <https://doi.org/10.1007/s40747-021-00336-7>.
- [33] Thomas Kailath. 1967. The divergence and bhattacharyya distance measures in signal selection. *IEEE Transactions On Communication Technology*, 15, 52–60.
- [34] Kai Shu, Deepak Mahudeswaran, Suhang Wang, Dongwon Lee, and Huan Liu. 2019. Fakenewsnet: a data repository with news content, social context and spatiotemporal information for studying fake news on social media. (2019). <https://arxiv.org/abs/1809.01286> arXiv: 1809.01286 [cs. SI].
- [35] Haibo He and Eduardo A. Garcia. 2009. Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21, (Sept. 2009), 1263–1284. <http://ieeexplore.ieee.org/document/5128907>.
- [36] Mohammad Q. Alnabhan and Paula Branco. 2024. BERTGuard: two-tiered multi-domain fake news detection with class imbalance mitigation. *Big Data and Cognitive Computing*, 8, 8, 93. doi:10.3390/bdcc8080093.
- [37] Kai Shu, Amy Sliva, Suhang Wang, Jiliang Tang, and Huan Liu. 2017. Fake news detection on social media: a data mining perspective. *ACM SIGKDD Explorations*, 19, 1, 22–36. doi:10.1145/3137597.3137600.
- [38] Suhaib Kh. Hamed, Mohd Juzaidin Ab Aziz, and Mohd Ridwan Yaakub. 2023. Fake news detection model on social media by leveraging sentiment analysis of news content and emotion analysis of users’ comments. *Sensors*, 23, 4, 1748. doi:10.3390/s23041748.
- [39] Thapanun Wongvorachan, Shiyu He, and Okan Bulut. 2023. A comparison of undersampling, oversampling, and SMOTE methods for dealing with imbalanced classification in educational data mining. *Information*, 14, 1. doi:10.3390/info14010054.
- [40] Lixuan Ding, Lanting Ding, and Richard O. Sinnott. 2020. Fake news classification of social media through sentiment analysis. In *Big Data – BigData 2020*. Surya Nepal, Wenqi Cao, Aziz Nasridinov, MD Zakirul Alam Bhuiyan, Xuan Guo, and Liang-Jie Zhang, (Eds.) Springer International Publishing, Cham, 52–67. ISBN: 978-3-030-59612-5.
- [41] Pouya Shaeri and Ali Katanforoush. 2023. A semi-supervised fake news detection using sentiment encoding and lstm with self-attention. In *2023 13th International Conference on Computer and Knowledge Engineering (ICCKE)*, 590–595. doi:10.1109/ICCKE60553.2023.10326287.